Exponentials & Logarithms

Worked Example Set

Chapter 3G–3L : Year 10 Mathematical Methods : 2026

Name: _____ Class: _____

How to use this booklet

Every section from 3G to 3L runs in the same order.

KEY IDEAS The rule for the section, all in one place. Come back to it whenever you are stuck.

I DO Fully worked examples. Your teacher talks through each line. The grey notes on the right say *why* each step happens.

WE DO Examples we finish together. Fill in each blank as the class works it out.

FIX IT A student's working with one mistake in it. Find the mistake, explain it, then fix it.

YOU DO Questions to try on your own, with space to write.

Red writing in a table is a worked first row. It shows you what to do in the rows below.

Sections 3K and 3L are marked 10A: they are extension. In every section, write the rule before you calculate, and answer in words when the question has a context.

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Sections follow *Essential Mathematics for the Victorian Curriculum 10 & 10A*, 3rd ed., Chapter 3. Every example and question in this booklet is original material written for this course.

3G Exponential growth and decay CORE

Key ideas

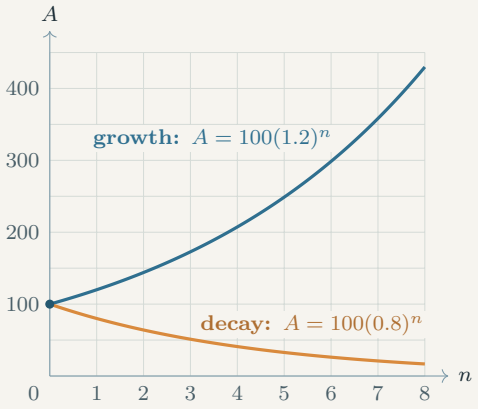
Growth: $A = A_0\left(1 + \dfrac{r}{100}\right)^n$

Decay: $A = A_0\left(1 - \dfrac{r}{100}\right)^n$

A_0 is the starting amount, r is the percentage change each time period, and n is the number of time periods.

The question says ...	Multiply by
grows by 20% each hour	1.2
increases by 3.5% each year	1.035
loses 15% each year	0.85
decreases by 2% each month	0.98

Multiplier more than 1: **growth**.
Multiplier between 0 and 1: **decay**.



Both start at 100. Growth multiplies by 1.2 each period; decay multiplies by 0.8.

I DO Example 1 — A growing colony

A colony of 500 bacteria increases by 20% every hour. Write a rule for the number of bacteria, N, after n hours. Then find the number of bacteria after 6 hours.

Solution

$\text{multiplier} = 1 + \frac{20}{100} = 1.2$	increases, so add the rate to 1
$N = 500(1.2)^n$	the rule
$N = 500(1.2)^6$	substitute $n = 6$
$= 500 \times 2.985984 = 1492.992$	calculate
≈ 1493	a whole number of bacteria

Answer. After 6 hours there are about 1493 bacteria.

Why not $500 + 6 \times 100$**?** The 20% is taken of the *new* amount each hour, so each increase is bigger than the last: $500 \to 600 \to 720 \to 864 \to \cdots$, which is +100, +120, +144, ... Adding the same amount each time is linear. Multiplying by the same number each time is exponential.

Notes

I DO **Example 2 — Medicine leaving the body**

A patient takes a 40 mg dose of a medicine. Each hour, the body removes 15% of the medicine that is still present.

(a) Write a rule for the amount of medicine, M mg, after t hours.

(b) Find the amount left after 4 hours, correct to one decimal place.

(c) After 4 hours, is more or less than half of the dose left?

Solution

(a) multiplier $= 1 - \frac{15}{100} = 0.85$ — removes 15%, so 85% stays

$M = 40(0.85)^t$ — the rule

(b) $M = 40(0.85)^4 = 40 \times 0.52200625$ — substitute $t = 4$

$= 20.88\ldots \approx 20.9$ — one decimal place

(c) $\frac{1}{2} \times 40 = 20$ and $20.9 > 20$ — compare with half the dose

Answer. (a) $M = 40(0.85)^t$ (b) about 20.9 mg (c) just **more** than half of the dose is left.

The multiplier is what stays. Losing 15% means keeping 85%, so write 0.85, never 0.15.

WE DO **Example 3 — What does the rule tell us?**

For each rule, write the starting amount, whether it shows growth or decay, and the percentage rate for each time period. The first row is done for you.

Rule	Starting amount	Growth or decay?	Rate each period
$V = 12\,000(0.92)^t$	12 000	decay, because $0.92 < 1$	$1 - 0.92 = 0.08$, so 8%
$P = 350(1.045)^n$			
$A = 60(2)^n$			
$M = 2.4(0.7)^t$			

Say it, for the rule for M:

"The rule starts at _____ and _____ by _____% each _____."

Notes

WE DO **Example 4 — When does the town drop below 6000?**

A town of 8000 people is shrinking by 5% each year. After how many whole years will the population first be below 6000?

Work through it together

Step 1. The multiplier is _____, so the rule is $P =$ _____.

Step 2. Try values of n. The first row is done for you.

Years, n	**Working**	**Population** P (nearest whole)	**Below 6000?**
3	$8000(0.95)^3 = 6859$	6859	No
4			
5			
6			

Step 3. The population first drops below 6000 after _____ years.

Looking ahead: in Section 3L you will solve $0.95^n = 0.75$ exactly with logarithms, and get $n \approx 5.61$.

FIX IT **Spot the error**

A car worth \$20 000 loses 15% of its value each year. Find its value after 1 year.

A student wrote:

$V = 20\,000(0.15)^n$, so after 1 year $V = 20\,000 \times 0.15 = \$3000$.

Explain the mistake. Then write the correct working.

Notes

YOU DO **Your turn — Section 3G** (show all working)

Question 3G.1

A painting bought for \$1200 increases in value by 4% each year.

(a) Write a rule for its value, V dollars, after n years.

(b) Find its value after 5 years, correct to the nearest cent.

Question 3G.2

A machine bought for \$65 000 loses 10% of its value each year. Find its value after 4 years.

Question 3G.3

Write the multiplier for each change.

(a) grows by 2.5% each year _____

(b) loses 2.5% each year _____

(c) doubles every hour _____

Question 3G.4

The rule $A = 750(0.94)^t$ gives an amount after t years.

(a) What is the starting amount? _____

(b) Is this growth or decay? _____

(c) What is the percentage rate each year? _____

Question 3G.5

An investment of \$2000 grows by 8% each year. Try values of n to find after how many whole years it is first worth more than \$3000.

3H Compound interest CORE

Key ideas

Compound interest

$$A = P\left(1 + \frac{r}{100}\right)^n \qquad \text{interest } I = A - P$$

P is the principal (the starting money), r is the rate **per period**, and n is the number of **periods**.

Simple interest, for comparison

$$I = \frac{P \times r \times t}{100}$$

The same interest every year. Round money to the nearest cent, at the **end** only.

Compounded	Rate per period	Number of periods in t years
annually	r	t
half-yearly	$r \div 2$	$2t$
quarterly	$r \div 4$	$4t$
monthly	$r \div 12$	$12t$

Divide the rate by the number of periods in a year. Multiply the years by the same number.

I DO Example 5 — Compounded annually

\$6000 is invested at 5% p.a., compounded annually, for 4 years.

(a) Find the final amount.

(b) Find the interest earned.

Solution

$$P = 6000, \quad r = 5, \quad n = 4 \qquad \text{write } P, r, n \text{ first}$$

(a)
$$A = 6000\left(1 + \tfrac{5}{100}\right)^4 = 6000(1.05)^4 \qquad \text{substitute}$$
$$= 6000 \times 1.21550625 = 7293.0375 \qquad \text{calculate}$$
$$\approx 7293.04 \qquad \text{nearest cent}$$

(b)
$$I = 7293.04 - 6000 = 1293.04 \qquad \text{interest} = A - P$$

Answer. (a) The final amount is \$7293.04. (b) The interest earned is \$1293.04.

Keep the full calculator value. Rounding 1.05^4 to 1.22 first gives $6000 \times 1.22 = \$7320$, which is \$26.96 too much.

Notes

I DO **Example 6 — Compounded monthly**

$10 000 is invested at 6% p.a., compounded monthly, for 3 years. Find the final amount.

Solution

$$r = 6 \div 12 = 0.5 \qquad \text{rate per month}$$

$$n = 3 \times 12 = 36 \qquad \text{number of months}$$

$$A = 10\,000\left(1 + \tfrac{0.5}{100}\right)^{36} = 10\,000(1.005)^{36} \qquad \text{substitute}$$

$$\approx 11\,966.81 \qquad \text{nearest cent}$$

Answer. After 3 years the investment is worth $11 966.81.

Compare with compounding annually. $10\,000(1.06)^3 = \$11\,910.16$. Compounding monthly earns $56.65 more, because the interest starts earning interest sooner.

WE DO **Example 7 — How much more does compounding earn?**

$4000 is invested for 5 years at 7% p.a. How much more does compound interest (compounded annually) earn than simple interest?

Work through it together

Simple interest. $I = \dfrac{4000 \times 7 \times 5}{100} =$ _____, so the final amount is _____.

Compound interest. $A = 4000\left(\text{_____}\right)^5 =$ _____

Difference. _____ − _____ = _____

Answer. Compound interest earns $_____ more.

Think. Simple interest pays $280 every year. Why does compound interest pay more than $280 in each year after the first?

_____

Notes

WE DO **Example 8 — Which account pays more?**

Mai has \$8000 to invest for 3 years.
Account A pays 6% p.a., compounded annually.
Account B pays 5.9% p.a., compounded monthly.
Which account gives more money, and how much more?

Work through it together

Predict first. Circle one: Account A Account B

Account A. Multiplier _____, $n =$ _____, so $A =$ \$_____

Account B. Rate per month $= \frac{5.9}{12}\%$, number of months $n =$ _____

$$A = 8000\left(1 + \frac{5.9}{1200}\right)^{36} = \$_____$$

Answer. Account _____ gives \$_____ more.

Notice: the account with the *lower* rate can still win, because it compounds 12 times a year.

FIX IT **Spot the error**

\$10 000 is invested at 6% p.a., compounded monthly, for 3 years. Find the final amount.

A student wrote:

The monthly rate is 0.5%, so $A = 10\,000(1.005)^3 = \$10\,150.75$.

Explain the mistake. Then write the correct working.

YOU DO **Your turn — Section 3H** (show all working)

Question 3H.1

\$2500 is invested at 3% p.a., compounded annually, for 6 years. Find the interest earned.

Question 3H.2

$15 000 is invested at 4.8% p.a., compounded monthly, for 2 years. Find the final amount.

Question 3H.3

$7000 is invested at 8% p.a., compounded quarterly, for 3 years. Find the final amount.

Question 3H.4

$900 is invested at 5.2% p.a., compounded half-yearly, for t years. Write a rule for the final amount, A. You do not need to calculate it.

Question 3H.5

$12 000 is borrowed at 9% p.a., compounded annually. No repayments are made.

(a) How much is owed after 3 years?

(b) How much of this is interest?

3I Introducing logarithms CORE

Key ideas

$\log_a x = y$ means $a^y = x$

A logarithm asks: "a to **what power** gives x?" The log **is** the power.

Two anchors: $\log_a 1 = 0$, because $a^0 = 1$
$\log_a a = 1$, because $a^1 = a$

Calculator: the log button is base 10, so $\log 1000$ means $\log_{10} 1000 = 3$.

Index form	Log form	Say it
$2^3 = 8$	$\log_2 8 = 3$	"2 to the power 3 gives 8"
$10^2 = 100$	$\log_{10} 100 = 2$	"10 to the power 2 gives 100"

I DO Example 9 — Swap between the two forms

(a) Write in log form: $2^5 = 32$, $10^{-3} = 0.001$, $9^{\frac{1}{2}} = 3$

(b) Write in index form: $\log_3 81 = 4$, $\log_5 0.04 = -2$, $\log_8 2 = \frac{1}{3}$

Solution

(a)

$$2^5 = 32 \Rightarrow \log_2 32 = 5$$
$$10^{-3} = 0.001 \Rightarrow \log_{10} 0.001 = -3$$
$$9^{\frac{1}{2}} = 3 \Rightarrow \log_9 3 = \frac{1}{2}$$

(b)

$$\log_3 81 = 4 \Rightarrow 3^4 = 81$$
$$\log_5 0.04 = -2 \Rightarrow 5^{-2} = 0.04$$
$$\log_8 2 = \frac{1}{3} \Rightarrow 8^{\frac{1}{3}} = 2$$

The move. The base stays the base. The power on one side is the answer on the other side.

I DO Example 10 — Ask "what power?"

Evaluate without a calculator.

(a) $\log_2 64$

(b) $\log_7 1$

(c) $\log_4 \dfrac{1}{16}$

(d) $\log_{27} 3$

Solution

(a)	$2^6 = 64$,	so $\log_2 64 = 6$	2 to what power gives 64?
(b)	$7^0 = 1$,	so $\log_7 1 = 0$	any base to the power 0 is 1
(c)	$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$,	so $\log_4 \frac{1}{16} = -2$	a fraction needs a negative power
(d)	$27^{\frac{1}{3}} = \sqrt[3]{27} = 3$,	so $\log_{27} 3 = \frac{1}{3}$	smaller than the base: a root

Patterns to look for. The answer is a fraction like $\frac{1}{16}$: a **negative** power. The answer is smaller than the base, like 3 from 27: a **fraction** power. The answer is 1: the log is **0**, whatever the base.

Notes

WE DO **Example 11 — Rewrite in index form, then solve**

Find the value of x. Write each equation in index form first.

Work through it together

(a) $\log_2 x = 5$ index form: _____ so $x =$ _____

(b) $\log_x 49 = 2$ index form: _____ so $x =$ _____

(c) $\log_{10} x = -2$ index form: _____ so $x =$ _____

(d) $\log_3 27 = x$ index form: _____ so $x =$ _____

In (b), $x = -7$ also gives $x^2 = 49$, but the base of a logarithm must be positive.

WE DO **Example 12 — Between which two whole numbers?**

Without a calculator, find the two whole numbers that each logarithm lies between.

Work through it together

(a) $\log_2 20$: _____ $< 20 <$ _____, so $\log_2 20$ is between _____ and _____.

(b) $\log_{10} 350$: _____ $< 350 <$ _____, so $\log_{10} 350$ is between _____ and _____.

Check (b) on the calculator: $\log 350 =$ _____

(c) $\log_3 100$: _____ $< 100 <$ _____, so $\log_3 100$ is between _____ and _____.

(a) and (c) are not base 10, so the log button cannot find them directly. Section 3L shows how — and this estimate is how you will check that answer.

FIX IT **Spot the error**

Evaluate $\log_2 8$.

A student wrote:

$\log_2 8 = 4$, because $8 \div 2 = 4$.

Explain the mistake. Then write the correct working.

Notes

YOU DO **Your turn — Section 3I** (show all working)

Question 3I.1

Write in log form.

(a) $6^3 = 216$

(b) $10^4 = 10\,000$

(c) $4^{-1} = 0.25$

Question 3I.2

Write in index form.

(a) $\log_2 128 = 7$

(b) $\log_5 1 = 0$

(c) $\log_{16} 4 = \frac{1}{2}$

Question 3I.3

Evaluate without a calculator.

(a) $\log_5 125$

(b) $\log_{10} 0.0001$

(c) $\log_9 3$

(d) $\log_2 \dfrac{1}{8}$

Question 3I.4

Solve for x.

(a) $\log_x 64 = 3$

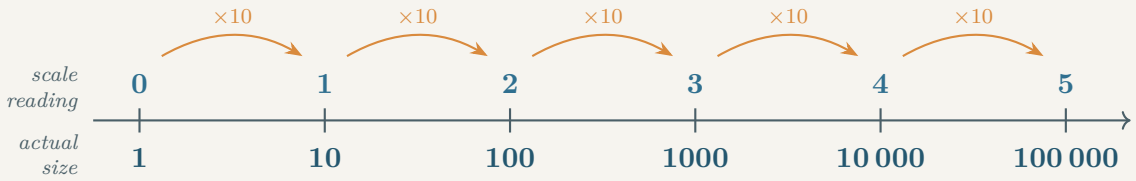
(b) $\log_4 x = 3$

Question 3I.5

Between which two whole numbers does $\log_2 50$ lie? Explain how you know.

3J Logarithmic scales CORE

Key ideas



One step on a log scale is $\times 10$ in real life. Two readings k apart: the actual sizes differ by a factor of 10^k. **Subtract** the readings, then raise 10 to that power.

Scale	One step means
Richter magnitude	$\times 10$ amplitude of shaking
pH (a **lower** pH is **more** acidic)	$\times 10$ acidity
decibels (dB)	every **10 dB** is $\times 10$ intensity

I DO Example 13 — Comparing earthquakes

(a) How many times larger is the amplitude of a magnitude 7.3 earthquake than the amplitude of a magnitude 4.3 earthquake?

(b) How many times larger is the amplitude of a magnitude 6.5 earthquake than the amplitude of a magnitude 5.0 earthquake?

Solution

(a) $7.3 - 4.3 = 3$ difference in readings

$10^3 = 1000$ each step is $\times 10$

(b) $6.5 - 5.0 = 1.5$ difference in readings

$10^{1.5} \approx 31.6$ between $10^1 = 10$ and $10^2 = 100$, as expected

Answer. (a) 1000 times larger. (b) About 32 times larger.

I DO Example 14 — How much more intense is the sound?

A lawnmower measures 90 dB. Normal conversation measures 60 dB. How many times more intense is the lawnmower's sound?

Solution

$90 - 60 = 30$ dB difference in readings

$30 \div 10 = 3$ steps one step is 10 dB

$10^3 = 1000$ each step is $\times 10$

Answer. The lawnmower's sound is 1000 times more intense.

Decibels are different. On the Richter and pH scales one step is 1. On the decibel scale one step is 10 dB, so the factor is $10^{\text{difference} \div 10}$. A 3 dB difference is $10^{0.3} \approx 2$: about twice as intense.

WE DO **Example 15 — Which is more acidic, and how many times?**

Remember: a **lower** pH is **more** acidic.

(a) Vinegar has pH 3. Black coffee has pH 5.

(b) Normal rain has pH 5.6. Seawater has pH 8.1.

Work through it together

(a) More acidic: _____ Steps apart, k: _____ $10^k =$ _____ times

(b) More acidic: _____ Steps apart, k: _____ $10^k \approx$ _____ times

WE DO **Example 16 — From "times bigger" back to a reading**

(a) An aftershock has $\frac{1}{100}$ of the amplitude of a magnitude 6.8 earthquake. What is the magnitude of the aftershock?

(b) An earthquake has 1000 times the amplitude of a magnitude 3.9 tremor. What is the magnitude of the earthquake?

Work through it together

(a) $\frac{1}{100} = 10^k$ with $k =$ _____, so the reading moves _____ steps _____ (up or down).

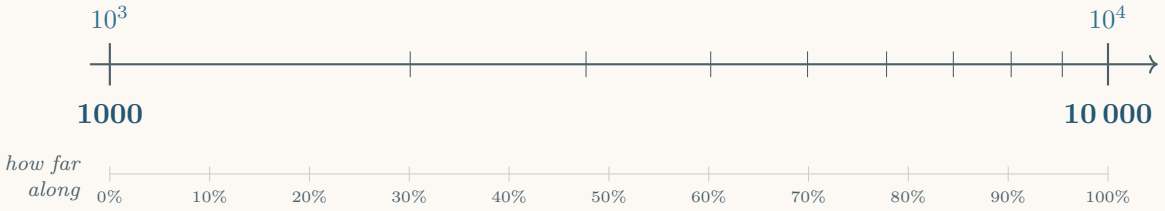
Magnitude of the aftershock = _____

(b) $1000 = 10^k$ with $k =$ _____, so the reading moves _____ steps _____ (up or down).

Magnitude of the earthquake = _____

WE DO **Example 17 — Where does 5000 go on a log axis?**

The axis below is a log axis from 1000 to 10 000. The eight small ticks are 2000, 3000, . . . , 9000.



Work through it together

(a) $\log 1000 = 3$ and $\log 10\,000 = 4$. Use a calculator: $\log 5000 =$ _____

(b) So 5000 sits _____% of the way from 1000 to 10 000. Mark 5000 on the axis.

(c) The number exactly halfway along is $10^{3.5} =$ _____. Mark it on the axis.

(d) Label the small ticks 2000 to 9000. What do you notice about how they are spaced?

_____

FIX IT **Spot the error**

Compare the amplitude of a magnitude 6 earthquake with the amplitude of a magnitude 3 earthquake.

A student wrote:

The magnitude 6 earthquake is twice as big, because $6 = 2 \times 3$.

Explain the mistake. Then write the correct working.

YOU DO **Your turn — Section 3J** (show all working)

Question 3J.1

How many times larger is the amplitude of a magnitude 8.1 earthquake than the amplitude of a magnitude 5.1 earthquake?

Question 3J.2

How many times larger is the amplitude of a magnitude 5.7 earthquake than the amplitude of a magnitude 5.2 earthquake?

Question 3J.3

A rock concert measures 110 dB. Busy traffic measures 80 dB. How many times more intense is the sound of the rock concert?

Question 3J.4

Solution A has pH 4. Solution B has pH 7.

(a) Which solution is more acidic?

(b) How many times more acidic is it?

Question 3J.5

A sound is 100 times more intense than a 50 dB sound. What is its level in decibels?

Question 3J.6

On a log axis from 100 to 1000, is 300 before or after the halfway point? Use $\log 300$ to explain.

3K Laws of logarithms 10A

Key ideas — each log law is an index law in disguise

Index law	Log law	In words
$a^m \times a^n = a^{m+n}$	**Product:** $\log_a x + \log_a y = \log_a(xy)$	add logs: multiply inside
$a^m \div a^n = a^{m-n}$	**Quotient:** $\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$	subtract logs: divide inside
$(a^m)^n = a^{mn}$	**Power:** $n\log_a x = \log_a(x^n)$	number in front: power inside

Special values: $\log_a 1 = 0$ $\log_a a = 1$ $\log_a \frac{1}{x} = -\log_a x$

Only when every log has the **same base**. There is **no** law for $\log_a(x+y)$.

I DO Example 18 — Write as a single logarithm

(a) $\log_3 5 + \log_3 4$

(b) $\log_2 30 - \log_2 6$

(c) $2\log_5 3$

Solution

(a) $\log_3 5 + \log_3 4 = \log_3(5 \times 4) = \log_3 20$ product law

(b) $\log_2 30 - \log_2 6 = \log_2\left(\frac{30}{6}\right) = \log_2 5$ quotient law

(c) $2\log_5 3 = \log_5\left(3^2\right) = \log_5 9$ power law

Write the law on every line. It shows your method, and it makes you check that the law fits before you use it. "Write as a single log" does not mean "evaluate": $\log_3 20$ is a finished answer.

I DO Example 19 — Combine first, then evaluate

Evaluate.

(a) $\log_6 4 + \log_6 9$

(b) $\log_3 54 - \log_3 2$

(c) $2\log_{10} 5 + \log_{10} 8 - \log_{10} 2$

Solution

(a) $\log_6 4 + \log_6 9 = \log_6 36 = 2$ product law, then $6^2 = 36$

(b) $\log_3 54 - \log_3 2 = \log_3 27 = 3$ quotient law, then $3^3 = 27$

(c) $2\log_{10} 5 + \log_{10} 8 - \log_{10} 2 = \log_{10} 25 + \log_{10} 8 - \log_{10} 2$ power law first

$= \log_{10}\left(\frac{25\times 8}{2}\right) = \log_{10} 100$ product and quotient laws

$= 2$ because $10^2 = 100$

Notes

WE DO **Example 20 — Use two known logs to find others**

Given $\log_{10} 2 \approx 0.301$ and $\log_{10} 3 \approx 0.477$, find:

(a) $\log_{10} 6$

(b) $\log_{10} 12$

(c) $\log_{10} 1.5$

Work through it together

(a) $6 = 2\times$ _____, so $\log 6 = \log 2 +$ _____ $\approx$ _____

(b) $12 = 2^2\times$ _____, so $\log 12 =$ _____ $\approx$ _____

(c) $1.5 = 3 \div$ _____, so $\log 1.5 =$ _____ $\approx$ _____

Check (b) on the calculator: $\log 12 =$ _____

WE DO **Example 21 — Fractions, halves and matching bases**

Evaluate.

(a) $\log_2 \dfrac{1}{32}$

(b) $\frac{1}{2}\log_3 81$

(c) $\log_4 8 + \log_4 2$

Work through it together

(a) Law or special value used: _____ Answer: _____

(b) Law or special value used: _____ Answer: _____

(c) Law or special value used: _____ Answer: _____

Notice (c): neither $\log_4 8$ nor $\log_4 2$ is a whole number, but together they are. Combine *before* you evaluate.

FIX IT **Spot the error**

Evaluate $\log_2(8+8)$.

A student wrote:

$\log_2(8+8) = \log_2 8 + \log_2 8 = 3 + 3 = 6$

Explain the mistake. Then write the correct working.

YOU DO **Your turn — Section 3K** (show all working)

Question 3K.1

Write as a single logarithm.

(a) $\log_7 3 + \log_7 5$

(b) $\log_5 36 - \log_5 4$

(c) $3\log_2 5$

Question 3K.2

Evaluate. Name the law you use in each part.

(a) $\log_{10} 50 + \log_{10} 20$

(b) $\log_7 98 - \log_7 2$

(c) $3\log_2 4$

(d) $\log_3 \dfrac{1}{9}$

Question 3K.3

Evaluate $2\log_6 3 + \log_6 4$.

Question 3K.4

Given $\log_{10} 2 \approx 0.301$ and $\log_{10} 3 \approx 0.477$, find:

(a) $\log_{10} 18$

(b) $\log_{10} 4.5$

3L Solving exponential equations using logarithms 10A

Key ideas — undo a power with a log

$$a^x = b \quad \Rightarrow \quad x = \log_a b = \frac{\log b}{\log a}$$

The fraction is how you get $\log_a b$ from the base-10 log button.

Calculator: for $\log_2 20$ type log(20) ÷ log(2) — not log(20 ÷ 2).

The four moves

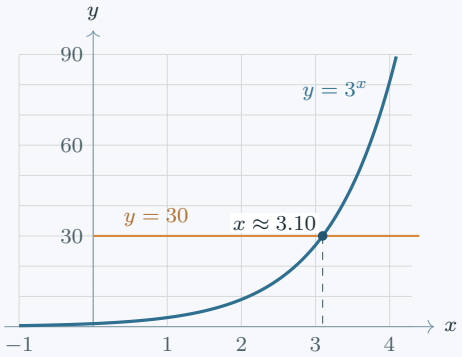
1. Get the power on its own: $a^x = b$.
2. Write the exact answer: $x = \log_a b$.
3. Calculate: $\log b \div \log a$.
4. Check: x sits between the two whole powers either side of b.

I DO Example 22 — The basic case

Solve $3^x = 30$. Give the exact answer and the answer correct to two decimal places.

Solution

$$\begin{aligned} x &= \log_3 30 && \text{exact answer} \\ &= \frac{\log 30}{\log 3} && \text{change to base 10} \\ &= \frac{1.4771\ldots}{0.4771\ldots} && \text{calculate} \\ &\approx 3.10 && \text{two decimal places} \end{aligned}$$



Check: $3^3 = 27 < 30 < 81 = 3^4$, and 30 is close to 27, so x is just above 3.

I DO Example 23 — Get the power on its own

Solve $5 \times 2^x = 70$, correct to two decimal places.

Solution

$$\begin{aligned} 2^x &= 14 && \text{divide both sides by 5} \\ x &= \log_2 14 = \frac{\log 14}{\log 2} && \text{exact answer} \\ &\approx 3.81 && \text{two decimal places} \end{aligned}$$

Check: $2^3 = 8 < 14 < 16 = 2^4$, and 14 is close to 16, so x is just under 4.

Don't multiply first. 5×2^x is **not** 10^x. Powers come before multiplication, so only the 2 is raised to the power x.

Notes

I DO **Example 24 — How long until the investment reaches \$8000?**

\$5000 is invested at 6% p.a., compounded annually. After how many years will it first be worth at least \$8000?

Solution

$$5000(1.06)^n = 8000 \qquad \text{set up the equation}$$
$$1.06^n = 1.6 \qquad \text{divide both sides by 5000}$$
$$n = \frac{\log 1.6}{\log 1.06} \approx 8.07 \qquad \text{use logs}$$

Check whole years: after 8 years the amount is $5000(1.06)^8 = \$7969.24$, which is not yet \$8000. After 9 years it is $5000(1.06)^9 = \$8447.39$.

Answer. The investment is first worth at least \$8000 after 9 years.

Round to suit the context. Interest is added only at the end of each year, so the balance jumps from \$7969.24 straight to \$8447.39. It is never exactly \$8000, so round **up** to the next whole year. Rounding 8.07 down to 8 gives the wrong answer.

WE DO **Example 25 — When is the car worth less than \$10 000?**

A car bought for \$35 000 loses 15% of its value each year. After how many years will it first be worth less than \$10 000?

Work through it together

Equation. $35\,000\left(_____\right)^n = 10\,000$

Power on its own. $0.85^n = \dfrac{10\,000}{35\,000} = _____$

Use logs. $n = \dfrac{\log \frac{2}{7}}{\log 0.85} \approx _____$

Check whole years. After 7 years: \$_____ After 8 years: \$_____

Answer. The car is first worth less than \$10 000 after _____ years.

Both $\log \frac{2}{7}$ and $\log 0.85$ are negative (the numbers are less than 1), so n comes out positive.

WE DO **Example 26 — A sum in the power**

Solve $2^{x+1} = 50$, correct to two decimal places.

Work through it together

Treat $x+1$ as the whole power. $x + 1 = \dfrac{\log _____}{\log _____} \approx _____$

Subtract 1. $x \approx _____$

Check. $2^5 = _____ < 50 < _____ = 2^6$, so $x+1$ is between 5 and 6.

Keep the full calculator value until the last line. Early rounding can change the second decimal place.

I DO **Example 27 — Stretch — an unknown in the power on both sides**

Solve $3^x = 2^{x+1}$. Give an exact answer and the answer correct to two decimal places.

Solution

$$\begin{aligned}
\log 3^x &= \log 2^{x+1} && \text{take the log of both sides}\\
x\log 3 &= (x+1)\log 2 && \text{power law brings the powers down}\\
x\log 3 &= x\log 2 + \log 2 && \text{expand}\\
x(\log 3 - \log 2) &= \log 2 && \text{collect the } x \text{ terms}\\
x &= \frac{\log 2}{\log 3 - \log 2} = \frac{\log 2}{\log 1.5} && \text{exact answer (quotient law underneath)}\\
&\approx 1.71 && \text{two decimal places}
\end{aligned}$$

Check: $3^{1.71} \approx 6.54$ and $2^{2.71} \approx 6.54$.

Why this works. The two sides have different bases, so you cannot just write $x = \log_a b$. Taking the log of both sides lets the power law bring x down to where you can collect it.

FIX IT **Spot the error**

Solve $2^x = 8$ using logarithms.

A student wrote:

$x = \dfrac{\log 8}{\log 2} = \log 4$, using the quotient law.

Explain the mistake. Then write the correct working.

YOU DO **Your turn — Section 3L** (show all working)

Question 3L.1

Solve, correct to two decimal places.

(a) $4^x = 100$

(b) $7^x = 3$

(c) $0.6^x = 0.1$

Question 3L.2

Solve $10 \times 1.5^x = 200$, correct to two decimal places.

Question 3L.3

A colony of 2000 bacteria grows by 20% each hour. After how many whole hours will there first be more than 50 000 bacteria?

Question 3L.4

A 120 g sample of a substance decays by 8% each day. After how many days is half of the sample left? Give your answer correct to one decimal place.

Question 3L.5

Stretch. Solve $5^x = 3^{x+2}$. Give an exact answer and the answer correct to two decimal places.

Mixed practice — which section? 3G–3L

In a test, nobody tells you which section a question comes from. For each question:

1. write the section (3G to 3L) that it comes from;
2. write the rule or law you will use;
3. solve it, and answer in words when there is a context.

Question M.1

Section: _____

Evaluate $\log_4 64$.

Question M.2

Section: _____

\$3000 is invested at 4% p.a., compounded annually, for 5 years. Find the final amount.

Question M.3

Section: _____

Solve $6^x = 50$, correct to two decimal places.

Question M.4

Section: _____

How many times larger is the amplitude of a magnitude 7.0 earthquake than the amplitude of a magnitude 4.5 earthquake?

Question M.5 Section: _____

Evaluate $\log_5 50 - \log_5 2$.

Question M.6 Section: _____

A town of 4000 people shrinks by 3% each year. Find the population after 10 years.